

# ANEESH GROVER

✉ [agrover@rockets.utoledo.edu](mailto:agrover@rockets.utoledo.edu) [LinkedIn](#) [GitHub](#) [Website](#)

## Experience

### AcademicAlly

July 2024 – August 2025

Founding Developer – ACM Thapar Society

React, Python, Pandas, Openpyxl, BeautifulSoup

- Served **10,000+** students by developing a responsive timetable portal for fast, reliable class schedule lookup.
- Built ETL pipelines to transform and validate **50,000+** Excel rows into optimized JSON datasets, implementing custom error handling that reduced parsing failures by **95%**.
- Supported deployment and maintenance of the **live portal**, coordinating with team members.

## Projects

### FocusBoard - Whiteboard Task Capture System

Kotlin, FastAPI, CRAFT, TrOCR, Phi-3 Mini, TFLite

- Built the Android app in Kotlin with ML Kit document scanning and an experimental **on-device YOLOv26n/TFLite** whiteboard detector, chosen for mobile size constraints.
- Engineered a **3-stage OCR pipeline** using CRAFT for line segmentation, TrOCR for transcription, and Phi-3 Mini for structured calendar JSON generation from handwritten input.
- Designed per-device request queues to serialize uploads, preventing concurrent TrOCR worker spawning and **eliminating cross-device race conditions** under simultaneous load.
- Returned structured JSON to the app for user approval before committing tasks to the shared E Ink display.

### Campus-Compute - Distributed Compute Framework

Python, FastAPI, WebSockets, Docker

- Architected a LAN-distributed compute system with a centralized coordinator and worker nodes, enabling automatic discovery and reconnect handling.
- Designed a capability-aware scheduler using CPU, RAM, and GPU metadata to optimize task placement across nodes.
- Built a real-time WebSocket dashboard for cluster telemetry, job orchestration, and containerized workload execution.

### folio.mcp - Personal MCP Server

TypeScript, Cloudflare Workers, Hono, MCP SDK

- Built a remote MCP server on Cloudflare Workers exposing GitHub activity, portfolio data, and an LLM-invocable contact tool.
- Added a KV layer for GitHub response caching with TTLs and IP-based rate limiting on the contact tool to prevent abuse.
- A simple companion frontend at [mcp.aneeshgrover.me](https://mcp.aneeshgrover.me)  with dynamic tool discovery; automated deploys via GitHub Actions.

### PatchIt - Civic Infrastructure Monitoring

PyTorch, YOLO, Depth-Anything-V2

- Co-led a crowdsourced platform with citizens and contractors to streamline reporting and resolution of road infrastructure issues.
- Built a YOLO-based pothole detection pipeline achieving **85.8% precision** and **81.7% mAP@50**, integrating depth estimation for 3D geometry and repair cost estimation.
- Runner-up (Open Innovation) at **NVIDIA SatHack Hackathon** among 1500+ participants.

## Skills

**Languages:** Python, Java, C++, JavaScript, TypeScript, SQL

**Software Engineering & Systems:** REST APIs, distributed systems, asynchronous processing, WebSockets, system design

**Machine Learning:** PyTorch, TensorFlow, transformers, vector databases, RAG

**Infrastructure & Tools:** Docker, Git, CI/CD, Supabase, OpenCV, CUDA

## Education

### University of Toledo

Toledo, OH

Bachelor of Science in Computer Science and Engineering

GPA: 3.5/4.0

## Leadership & Involvement

### Association for Computing Machinery (ACM)

Nov 2023 – Present

Executive Board Member & Machine Learning Bootcamp Lead

- Organized Hacklipse 2024, ACM Thapar's flagship hackathon, coordinating operations for **300+** participants.
- Mentored **100+** students through ACM's Machine Learning Bootcamp by designing curriculum and leading weekly technical sessions.

## Achievements

- Secured **runner-up** in **Open Innovation Category** at **NVIDIA's SatHack**, competing against **1500+** participants.
- Awarded Best Use Case for AcademicAlly at ACM Projectathon 2024.